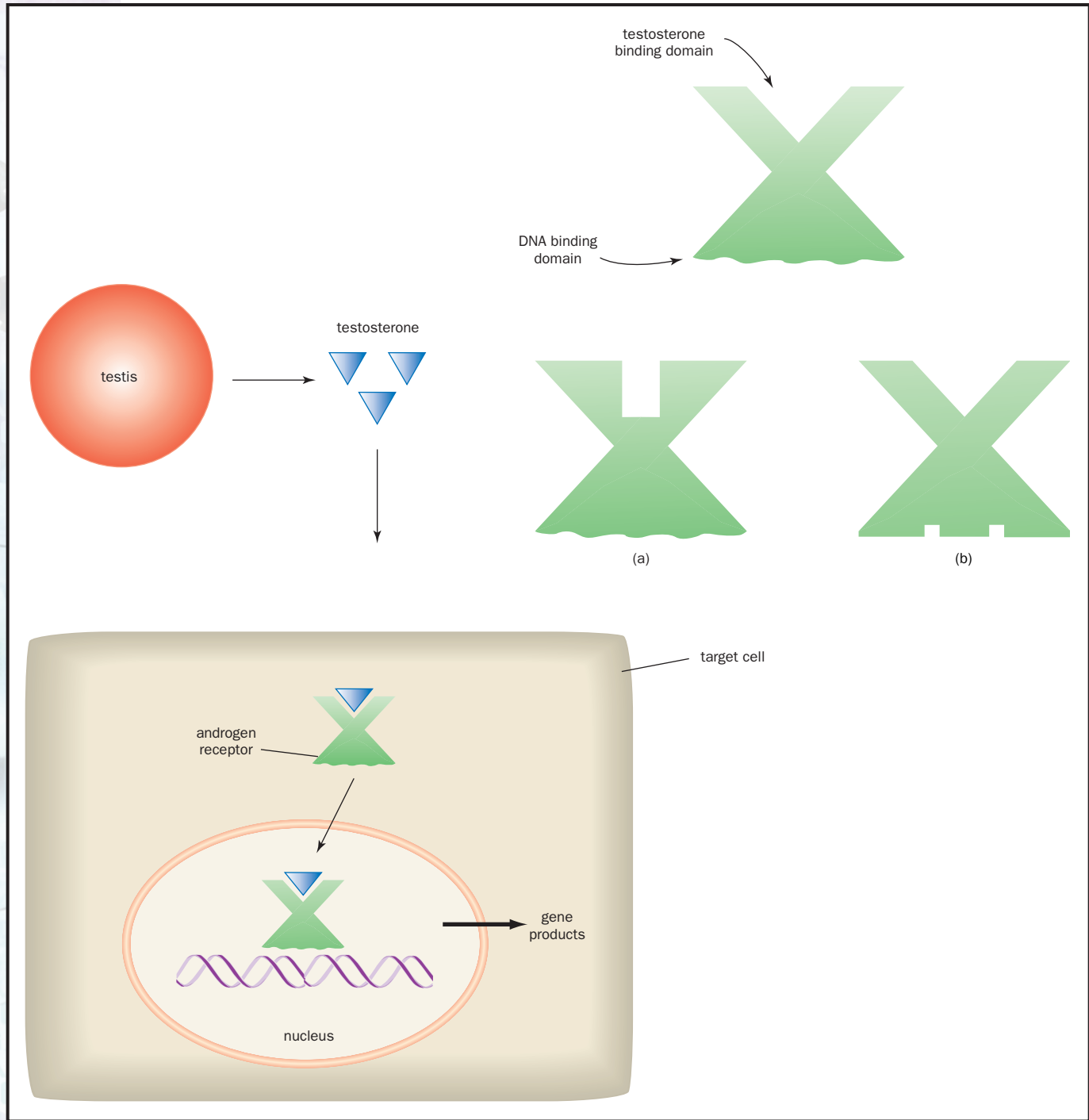




Testosterone from the testis binds to the androgen receptor. This complex then triggers gene expression. Mutation in the receptor prevents either testosterone binding (a), DNA binding (b), or some other activity.

long, though fewer than 3,000 of these actually code for amino acids in the protein. The protein formed from the gene has different **domains** that perform different functions. One region binds testosterone, another regulates the movement of the complex to the nucleus, and a third binds the complex to the DNA. Other regions, some overlapping, control other functions.

Mutations to the coding portion for any one of these domains can prevent the receptor complex from functioning properly. All known mutations

in the AR gene cause a loss of function and exhibit the recessive inheritance pattern. A male carries only one X chromosome, and receives only one copy of the AR gene. If this gene is mutated, the male will have androgen insensitivity syndrome.

Women with one mutated AR gene will not exhibit the syndrome but instead will be carriers, whose male children have a 50 percent risk of inheriting the mutant gene. Since affected individuals are sterile, they cannot pass it on to offspring. It is believed that about one-third of all cases are due to new mutations, which are not present in the mother's genes but which arise in the development of the early embryo. Genetic testing is available for women who desire children but who have a family history of androgen insensitivity. For women who are carriers, prenatal testing is available to determine if a fetus has inherited the mutant gene.

The Range of Androgen Insensitivity Syndromes

AIS occurs in a range of forms, from complete to mild. Most mutations to the testosterone-binding region, and some other types of mutations, cause complete androgen insensitivity syndrome (CAIS). In this form of the syndrome, the XY person is born phenotypically female, and from birth is raised as a girl. Gender identification (the internal sense of being male or female) is female. **Sexual orientation** is typically heterosexual, and so most CAIS individuals are attracted to males. At puberty, estrogen production by the adrenal glands causes breasts to develop. However, no pubic or armpit hair develops, since in males and females this is controlled by testosterone, and no menstruation occurs. It is at this point that the condition is usually diagnosed. Once discovered, the testes are usually surgically removed to prevent the possibility of testicular cancer, which is more common in people with CAIS. The woman is infertile, but may be able to enjoy sexual relations if the vagina is long enough to prevent pain during intercourse; or, if the vagina is not long enough, it can be surgically lengthened. CAIS is thought to occur in 2 to 5 births per 100,000.

Those with partial androgen insensitivity syndrome (PAIS) have androgen receptors that are partially responsive to testosterone, and a range of outcomes may result. A person with PAIS may be born with external genital structures that are not typically male or typically female, a condition called intersexuality. The appearance of the genitals may range from predominantly male to predominantly female. There may be a very small penis or enlarged clitoris, abnormalities in the location of the urethra, and partial fusion of the labia. Breasts may develop in males at puberty. Internal gender identification may be with either sex. PAIS is thought to be as common as CAIS.

Ambiguous genitals are often surgically altered at birth. Problems arise when the surgically assigned sex conflicts with the internal gender identification, which develops early and becomes even more pronounced through late childhood and puberty. Increasing understanding of PAIS and sensitivity to the issues of gender identification have brought new awareness about the potential for these problems, but the practice of surgical sex assignment is still common. In 2001 the British Association of Paediatric Surgeons recommended that surgery “only be undertaken with considerable caution and following full multidisciplinary investigation and counseling of the parents.”

domains regions

sexual orientation
attraction to one sex or
the other



Other therapies include hormone treatments and psychological counseling, including family counseling, and these are often part of the treatment, with or without surgery.

Kennedy Disease

Kennedy disease is a neurological condition that is also due to a mutation of the androgen receptor gene. Affected individuals are phenotypically normal males who are fertile, although after puberty they may develop enlarged breasts, consistent with very mild androgen insensitivity. The disorder causes progressive weakness over several decades, along with tremor, difficulty swallowing, and some sensory problems. The mutation that causes Kennedy disease is an expanded “triplet repeat” of CAG nucleotides, making this condition one of the family of triplet repeat diseases that includes Huntington’s disease. SEE ALSO APOPTOSIS; HORMONAL REGULATION; INHERITANCE PATTERNS; SEX DETERMINATION; SEXUAL ORIENTATION; TRANSCRIPTION FACTORS; TRIPLET REPEAT DISEASE; X CHROMOSOME; Y CHROMOSOME.

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Aneuploidy See *Chromosomal Aberrations*

Antibiotic Resistance

Antibiotic resistance is the ability of a bacterium or other microorganism to survive and reproduce in the presence of antibiotic doses that were previously thought effective against them. Examples of microbe resistance to antibiotics dot the countryside, plaguing humankind. For instance, in February 1994 dozens of students at La Quinta High School in southern California were exposed to the pathogenic (disease-causing) agent, *Mycobacterium tuberculosis*, and eleven were diagnosed with active tuberculosis. Many strains of this bacterium are multi-drug resistant (MDR). As for the sexually transmitted pathogen *Neisseria gonorrhea*, which causes gonorrhea, the antibiotics penicillin and tetracycline that were used against it in the 1980s can no longer be the first lines of defense, again because of antibiotic resistance. If only 2 percent of a *N. gonorrhea* population is antibiotic resistant, a community-wide infection of this persistent strain can occur.



Antibiotic resistant *Staphylococcus aureus* bacteria grow on this medium filled with antibiotics.

Mechanisms of Resistance

Antibiotics, whether made in the laboratory or in nature by other microbes, are designed to hinder metabolic processes such as cell wall synthesis, protein synthesis, or transcription, among others. If humans are to prosper against microbial disease, it is necessary to understand how and why bacteria are able to mount their clever defenses. Aided with the knowledge of the genetics and mechanisms of resistance, scientists can discover new ways to combat the resistant bacteria.

The phenomenon of antibiotic resistance in some cases is innate to the microbe. For instance, penicillin directly interferes with the synthesis of bacterial cell walls. Therefore, it is useless against many other microbes such as fungi, viruses, wall-less bacteria like *Mycoplasma* (which causes “walking pneumonia”), and even many **Gram negative** bacteria whose outer membrane prevents penicillin from penetrating them. Other bacteria change their “genetic programs,” which allows them to circumvent the antibiotic effect. These changes in the genetic programs can be in the form of chromosomal mutations, acquisition of R (resistance) **plasmids**, or through transposition of “**pathogenicity islands**.”

An example of a chromosomal mutation is the increasing number of cases of penicillin-resistant *Neisseria gonorrhoeae*. This bacterium mutated the gene coding for a porin protein in its outer membrane, thereby halting the transport of penicillin into the cell. This is also termed “vertical evolution,” meaning that the spread occurs through bacterial population growth. The most common method by which antibiotic resistance is acquired is through the **conjugation** transfer of R plasmids, also termed “horizontal evolution.” In this method the bacteria need not multiply to spread their plasmid. Instead the plasmid is moved during conjugation. These plasmids often code for resistance to several antibiotics at once.

The third method is transfer due to transposable elements on either side of a “pathogenicity island,” which is group of genes that appear on the DNA

Gram negative bacteria that do not take up Gram stain, due to membrane structure

plasmids small rings of DNA found in many bacteria

pathogenicity ability to cause disease

conjugation a type of DNA exchange between bacteria



and carry the codes for several factors which make the infection more successful. These transposable elements allow the genes to jump from bacteria to bacteria or simply from chromosome to plasmid within the organism.

The “road blocks” that bacteria have evolved which result in antibiotic resistance employ several mechanisms. One strategy is simply to destroy or limit the activity of the antibiotic. The beta-lactamases are enzymes which render the penicillin-like antibiotics dysfunctional by cleaving a vital part of the molecule. Some bacteria can deactivate antibiotics by adding chemical groups to them; for instance, by changing the electrical charge of the antibiotic through the addition of a phosphate group. Other bacteria accomplish a similar effect by bulking themselves up with the addition of an acetyl group.

Still other bacteria acquire resistance by simply not allowing the antibiotic to enter the cell. The bacterium mentioned above, *Neisseria gonorrhea*, has altered porin proteins, thereby stopping uptake of the antibiotic. Some bacteria acquire intricate pumping mechanisms to expel the drug when it gains entry to their cell.

Finally, bacteria may mutate the gene for the target **macromolecule** with which the antibiotic is supposed to bind. For example, tetracycline binds to and inhibits ribosomes, so a mutation in the ribosomal genes may cause conformational alterations in the ribosomal proteins that prevent tetracycline from binding but still allow the ribosome to function.

Resistance and Public Health

The effects of antibiotic resistance are reflected in the agriculture, food, medical, and pharmaceutical industries. Livestock are fed about half of the antibiotics manufactured in the United States as a preventative measure, rather than in the treatment of specific diseases. Such usage has resulted in hamburger meat that contains drug-resistant and difficult-to-treat *Salmonella Newport*, which has led to seventeen cases of gastroenteritis including one death. Some MDR-tuberculoid strains arise because patients are reluctant to follow the six-months or more of treatment needed to effectively cure tuberculosis. If the drug regimen is not followed, less sensitive bacteria have the chance to multiply and gradually emerge into resistant strains. In other cases the “shotgun” method of indiscriminately prescribing/taking several antibiotics runs the risk of creating “super MDR-germs.” Moreover, millions of antibiotic prescriptions are written by physicians each year for viral infections, against which antibiotics are useless. The patient insists on a prescription, and many doctors willingly go along with the request.

Because global travel is common, the potential of creating **pandemics** is looming. In many Third World countries, diluted antibiotics are sold on the black market. The dosage taken is often too low to be effective, or the patient takes the drug for a very short time. All these behaviors contribute to the development of resistant strains of infectious organisms. If humans are to gain the upper hand against MDR bacteria, it is the responsibility of these industries and the public to educate themselves and to engage in careful practices and therapy. SEE ALSO CONJUGATION; EUBACTERIA; MUTAGEN; PLASMID; TRANSPOSABLE GENETIC ELEMENTS.

Paul K. Small

macromolecule large molecule such as a protein, a carbohydrate, and a nucleic acid

pandemics diseases spread throughout entire populations

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Antisense Nucleotides

Antisense nucleotides are strings of RNA or DNA that are **complementary** to “sense” strands of nucleotides. They bind to and inactivate these sense strands. They have been used in research, and may become useful for therapy of certain diseases.

Antisense RNA

Messenger RNA (mRNA) is a single-stranded molecule used for protein production at the **ribosome**. Because its sequence is used for **translation**, mRNA is called a “sense” strand or sense sequence. A complementary sequence to that mRNA is an “antisense” sequence. For instance, if the mRNA sequence was AUGAAACCCGUG, the antisense strand would be UACUUUGGGCAC. Complementary sequences will pair up in RNA just as they do in DNA. When this happens to an mRNA, however, it can no longer be translated at the ribosome, no protein synthesis occurs, and the “duplex” RNA is degraded.

This phenomenon has been used experimentally and commercially to block the synthesis of specific proteins in transgenic organisms (ones to which a foreign gene has been added). The strategy is to add a synthetic gene that, when transcribed, will make the antisense RNA sequence for the target protein’s mRNA.

This technique was first used commercially in 1988 for the Flavr-Savr tomato. The gene chosen for inactivation was polygalacturonase (PG), whose enzyme unlinks pectins in the plant cell wall, thereby softening it. The intent was to increase the time the fruit could be left to ripen without softening, thus increasing flavor of commercial tomatoes. The Calgene company created a transgenic tomato plant expressing the antisense RNA for PG mRNA, and reduced PG production by up to 90 percent. Although the tomato was not a commercial success, it demonstrated the potential for this strategy.

Antisense RNA is currently being investigated as a human therapy for certain forms of cancer. The goal is to use gene therapy techniques to insert an antisense gene into tumor cells. Many cancers are due to overexpression of the genes that promote cell proliferation, called tumor suppressor genes. Antisense RNA might be able to inhibit this overexpression. Another target is the *BCL-2* gene, whose protein prevents apoptosis, or programmed cell death. In certain cancers, the *BCL-2* gene is overactive, preventing death of cells and leading to their proliferation. Antisense therapy against *BCL-2* is currently being tested under the trade name Genazyme.

complementary matching opposite, like hand and glove

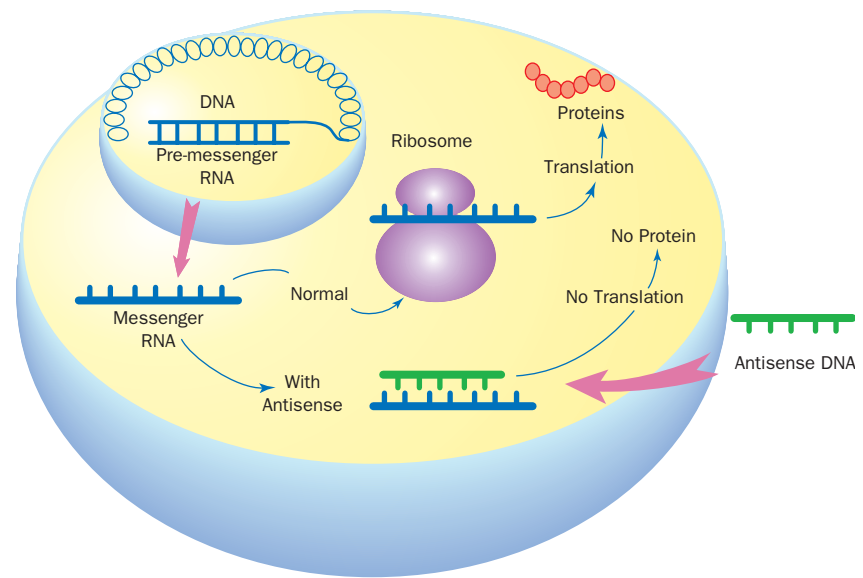
ribosome protein-RNA complex at which protein synthesis occurs

translation synthesis of protein using mRNA code

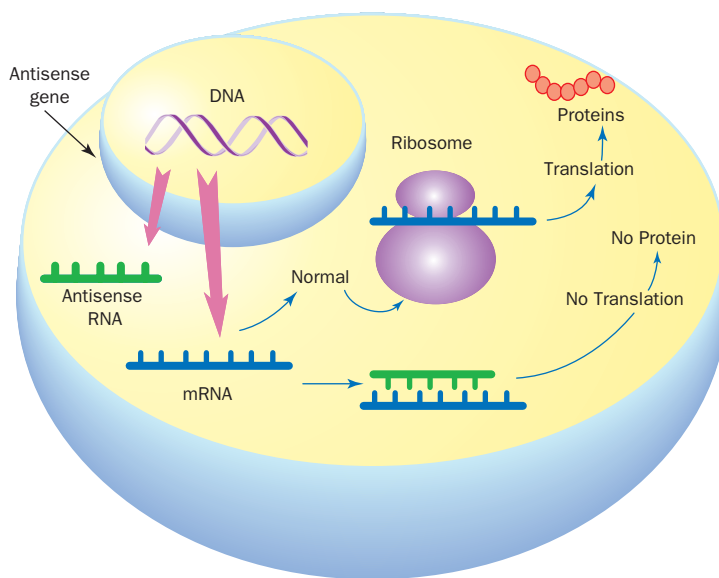


Antisense nucleotides bind to “sense” mRNA, preventing translation of the RNA message at the ribosome, thus preventing protein synthesis. Adapted from <<http://www.cem.msu.edu/~cem181h/projects/97/antisense/dia1.gif>>.

Antisense DNA



Antisense RNA



Antisense DNA

Antisense DNA strands can also be made (note that in the double helix, the side of the DNA that is transcribed is itself antisense). Short antisense strands of DNA can be introduced into cells, which then bind with target mRNA. Antisense DNA is currently an approved therapy for cytomegalovirus infections of the eye, under the trade name Vitravene. Vitravene targets two different viral proteins. Antisense DNA is also being explored for therapy of HIV, some cancers, and other diseases.

One advantage of using antisense therapy in treating infectious diseases such as virus infections is that it can be tailored to the particular strain in circulation, and then modified as the virus mutates. One difficulty in applying this therapy is successfully delivering the antisense DNA or RNA to all target tissues (for instance, making sure the antisense strands reach infected blood cells for HIV). Another problem is maintaining prolonged suppression of target protein expression, since the antisense molecule will eventually be degraded by the cell's **nuclease** enzymes. One strategy to prevent degradation is to chemically modify the DNA to interfere with nuclease action.

nuclease enzyme that cuts DNA or RNA

RNA Interference

Investigation of the mechanism of action of antisense RNA led to the surprising discovery that naturally occurring double-stranded RNA molecules (dsRNA) suppress gene expression as well as or better than antisense sequences. This suppression by dsRNA of expression of the related gene is called RNA interference. dsRNA molecules are cut into short segments by nucleases; the antisense strand of such a segment then peels off and binds with its complementary mRNA. This new, double-stranded RNA is then subject to further nuclease attack. RNA interference is believed to be an ancient means of protecting against double-stranded RNA viruses. Further understanding of RNA interference may lead to improvements in or replacement of antisense therapies. SEE ALSO GENE THERAPY; NUCLEASES; NUCLEOTIDE; RNA INTERFERENCE; TRANSGENIC PLANTS.

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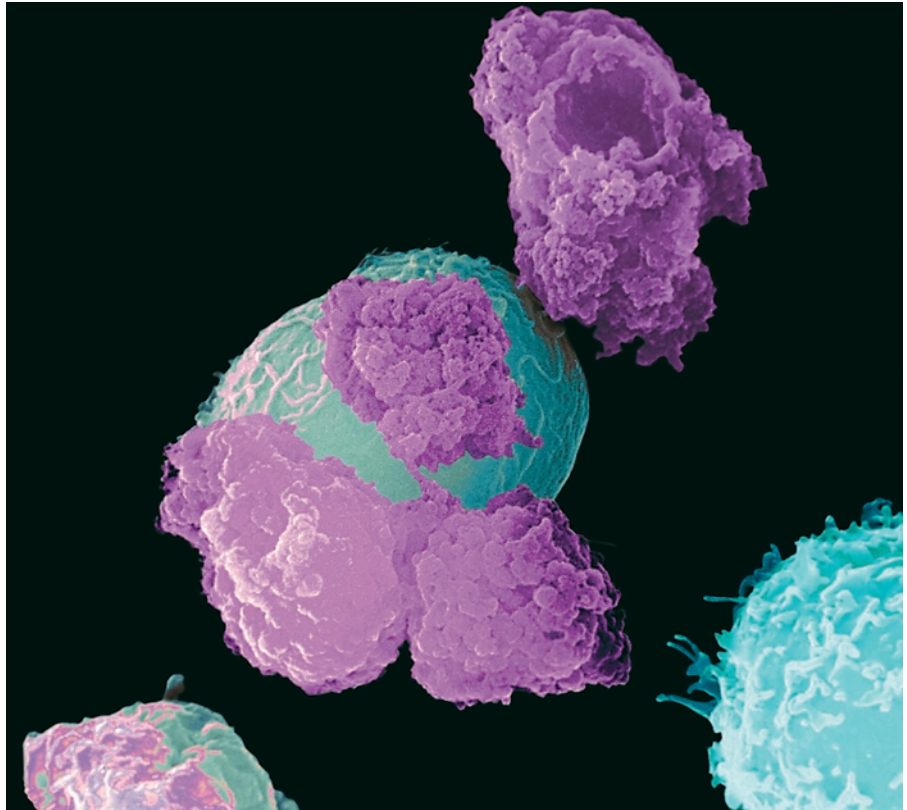
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Apoptosis

Death is an inevitable fact of life for organisms. Increasingly, biologists have come to realize that death is also, in many cases, an important and predestined fate of individual cells of organisms. Apoptosis is a process by which cells in a multicellular organism commit suicide. While cells can die as a result of **necrosis**, apoptosis is a form of death that the cell itself initiates, regulates, and executes using an elaborate arsenal of cellular and molecular machinery. For this reason, the term apoptosis is often used interchangeably with the term "programmed cell death," or PCD (although technically, apoptosis is but one particular form of programmed cell death). There is some disagreement on the origins of the word. The word apoptosis has ancient Greek origins, referring to the falling of leaves, or possibly "dropping of scabs" or "falling off of bones." There is even less agreement on its proper pronunciation, and even specialists in the field seem to use every possible way to say the word. "A-pop-TOE-sis" and "AP-oh-TOE-sis" are both common.

necrosis cell death from injury or disease

Magnification (6000×) of apoptosis, or cell death. Programmed cell death in the human body allows, among other things, for young children's brains to develop and for a female's monthly menstruation to occur.



APOPTOSIS GENES IN *C. ELEGANS*

Much of our understanding of what causes apoptosis comes from genetic studies in *Caenorhabditis elegans*. Several cell death proteins (CED) were identified in *C. elegans* by studying apoptosis-defective mutants. The main executioner is CED-3, a caspase, which becomes activated by CED-4, another caspase. The central guardian protecting cells against apoptosis is CED-9, which inhibits the actions of CED-4 and CED-3. CED-9 has a mammalian homolog, called BCL-2, which serves a similar role in mammals.

Why Cells Commit Suicide

Why do cells commit apoptosis? There seem to be two major reasons. First, apoptosis is one means by which a developing organism shapes its tissues and organs. For instance, a human fetus has webbed hands and feet early on its development. Later, apoptosis removes skin cells, revealing individual fingers and toes. A fetus's eyelids form an opening by the process of apoptosis. During metamorphosis, tadpoles lose their tails through apoptosis. In young children, apoptosis is involved in the processes that literally shape the connections between brain cells, and in mature females, apoptosis of cells in the uterus causes the uterine lining to slough off at each menstrual cycle.

Cells may also commit suicide in times of distress, for the good of the organism as a whole. For example, in the case of a viral infection, certain cells of the immune system, called cytotoxic T lymphocytes, bind to infected cells and trigger them to undergo apoptosis. Also, cells that have suffered damage to their DNA, which can make them prone to becoming cancerous, are induced to commit apoptosis.

The Regulatory Mechanism

The cellular mechanisms that regulate and cause apoptosis were first elucidated by genetic studies of the roundworm, *Caenorhabditis elegans*. Normally, in the development of a *C. elegans* worm, one out of every eight body cells produced is eliminated by programmed cell death. By studying mutants in which either too many or too few cells died, worm geneticists identified many of the proteins that control apoptosis. Subsequently, the critical med-

ical relevance of apoptosis became clear when biologists discovered that mammals contain many of the same genes that control apoptosis in worms. More strikingly, they found that many of these genes were mutated in tumors from cancer patients. Other genes often found to be mutated in cancers are those which regulate the cell cycle, which is the complex set of processes controlling how and when cells divide. These two findings led cancer researchers to recognize that cancer, a disease of uncontrolled cell proliferation, can result either from too much cell division or not enough apoptosis. Because of this important finding, apoptosis has become the subject of intense medical research, and molecules that regulate apoptosis are being studied as potential targets for anti-cancer drug therapies.

A cell can be triggered to undergo apoptosis either by external signaling molecules, such as so-called “death activator” proteins, or through molecules that reside within the cell and monitor events that might commit the cell to suicide, such as damage to DNA. There are several biochemical pathways that lead to apoptosis. One of the major pathways involves inducing **mitochondria** to leak one of their proteins, cytochrome c, into the cytosol. This in turn activates a set of related proteases (enzymes that degrade proteins) called caspases. Ultimately, the caspases degrade proteins in the cell and activate enzymes that degrade other cell constituents, such as the DNA. Cells undergoing apoptosis exhibit characteristic morphological and biochemical traits, which can be recognized by microscopic examination or biochemical assays. Apoptosis can occur in as little as twenty minutes, after which the cell “corpse” typically becomes engulfed and completely degraded by neighboring **phagocytic** cells that are present in the tissue and attracted to the apoptotic cell. SEE ALSO CANCER; CELL CYCLE; ROUNDWORM: *CAENORHABDITIS ELEGANS*; SIGNAL TRANSDUCTION.

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Arabidopsis thaliana

Arabidopsis thaliana, or thale cress, is a small flowering plant in the mustard family. *Arabidopsis* has no inherent agricultural value and is even considered a weed, but it is one of the favored model organisms of plant geneticists and molecular biologists, and it is the most thoroughly studied plant species at the molecular level. Model organisms have traits that make them attractive and convenient for biologists, who anticipate being able to extend their findings to other, less easily studied species. *Arabidopsis* is small and easy to grow, allowing researchers to cultivate it with minimal investments in effort and laboratory space. It has a short generation time, taking about six weeks for a seed to grow into a mature plant that produces more seeds. This rapid maturation

Caspase inhibitors are being investigated as a possible means to slow the progress of Huntington’s disease, a degenerative brain disease.

mitochondria energy-producing cell organelle

phagocytic cell-eating



The small genome and short life cycle of the *Arabidopsis thaliana* help to make it a useful tool for plant geneticists. Study of this plant's genome could help to shape development of the commercial mustard plant crops, and could provide useful information for study of other, more complex, flowering plants.

pollen male plant sexual organ

enables biologists to conduct genetic cross experiments in a relatively short period of time. A single mature plant can produce over 5,000 seeds, another property that makes *Arabidopsis* convenient for use in genetic analysis.

A Small and Simple Genome

Beyond these basic traits, other attributes of *Arabidopsis* make it particularly well-suited for analysis by modern molecular genetic methods. Its genome (the amount of DNA in each set of chromosomes) is only about 125 million base pairs. This is small compared to many other plants, and makes searching for particular genes easier in *Arabidopsis* than in plants with larger genomes. For comparison, the genome sizes for rice (*Oryza sativa*), wheat (*Triticum aestivum*), and corn (*Zea mays*) are 420 million, 16 billion, and 2.5 billion base pairs, respectively. Furthermore, the *Arabidopsis* genome is contained on just five pairs of chromosomes, making it easier for geneticists to locate specific genes.

Geneticists can carry out crosses (interbreeding two different plant strains) with *Arabidopsis* by introducing the **pollen** from one plant to the

stigma on another. This mode of reproduction, called outcrossing, is useful for combining mutations from different plants. Alternatively, *Arabidopsis* can reproduce by a process called selfing, in which an individual plant uses its own pollen to fertilize its **ovules**.

Selfing, which is not possible in many plants, is very useful for geneticists who wish to study mutations. Most mutations are recessive, which means that they physically manifest themselves (display a **phenotype**) only when they are present on the chromosomes contributed by both the ovule and the fertilizing pollen. In selfing, **heterozygous** mutations (which are present on only one of the two sets of chromosomes) will become **homozygous** (present on both) in one quarter of the progeny produced in this manner.

Arabidopsis and Transformation

Another property that endears *Arabidopsis* to plant molecular biologists is that it is easily transformed. Transformation is a method for introducing foreign DNA into an organism. This technique is invaluable for studying how genes function and interact with other genes. Biologists usually transform plants by infecting them with genetically engineered varieties of a bacterium, *Agrobacterium tumefaciens*. In nature, when *Agrobacterium* infects plants, it inserts certain genes directly into the plant cells, causing a disease called crown gall. The genetically engineered *Agrobacterium* strains have had their disease-causing genes removed. They can still infect a plant and insert their DNA, but do not cause a disease. To transform plants, the molecular biologist inserts the foreign gene to be studied into *Agrobacterium*, which will then transfer the gene to a plant that it infects. This transformation technique does not work well on many other plant species, limiting the utility of those plants for molecular genetic analysis.

Arabidopsis researchers also use a variation on the *Agrobacterium*-mediated transformation technique to introduce mutations in the plant. Studying the effect of a mutation in a particular gene often yields critical information about the normal function of that gene. Because *Agrobacterium* inserts its transforming DNA randomly in the genome, in many cases the DNA gets inserted directly within a gene sequence. This usually destroys the function of the disrupted gene, resulting in a “knockout mutant.” Furthermore, the piece of transformed DNA (T-DNA) that is inserted in the disrupted plant gene can serve as a flag for tracking down the gene by molecular biology methods. Large-scale projects using this T-DNA insertion technique are underway to mutate, identify, and characterize every gene in the *Arabidopsis* genome.

The First Completely Sequenced Plant Genome

At the end of 2000, an international team of researchers announced that *Arabidopsis* was the first plant to have its complete genome sequence—the exact order of essentially all 125 million DNA base pairs—determined. The project revealed that *Arabidopsis* contains over 25,500 genes. By identifying and studying these genes, biologists are learning lessons about plant biology that could provide important advances in agriculture, such as improved crop resistance to pathogens, salt, light stress, and drought, and to the production of more healthful edible oils, pharmaceuticals, and biodegradable plastics.

stigma female plant sexual organ

ovules eggs

phenotype observable characteristics of an organism

heterozygous characterized by possession of two different forms (alleles) of a particular gene

homozygous containing two identical copies of a particular gene



circadian relating to day or day length

phylogenetic related to the evolutionary development of a species

Arabidopsis research has also produced important discoveries in fundamental plant science, such as the identification of a plant hormone receptor, a clearer understanding of how plants sense and respond to light, and more about the processes that induce plants to form flowers. *Arabidopsis* research may even have direct relevance to human biology. For example, a photoreceptor protein that regulates **circadian** rhythms in *Arabidopsis* was found to share sequence similarity to a retinal photoreceptor, which may perform a similar role in mammals. SEE ALSO INHERITANCE PATTERNS; MODEL ORGANISMS; TRANSFORMATION; TRANSGENIC PLANTS.

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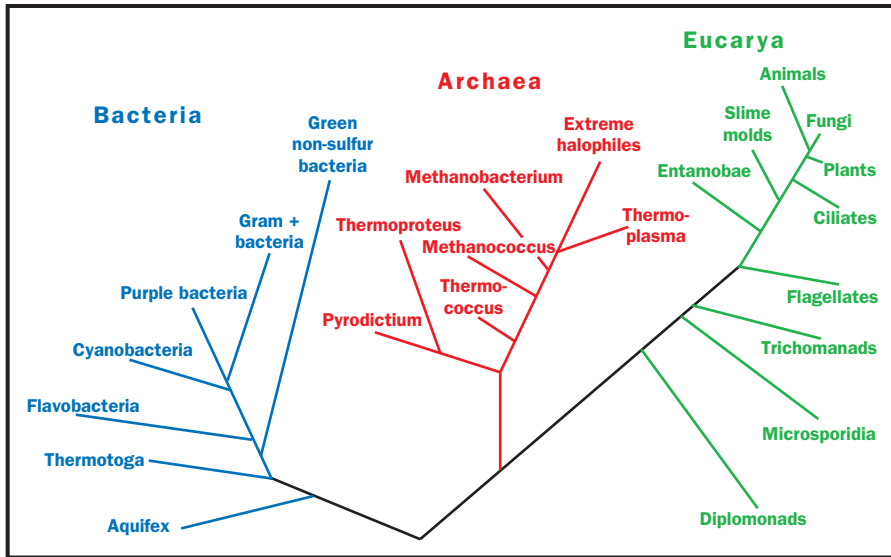
Archaea

Members of the Archaea comprise one of the three principal domains of living organisms in the universal **phylogenetic** tree of life. The other two principal domains are the Bacteria and the Eukarya. The phylogenetic tree is a theoretical representation of all living things, constructed on the basis of comparative ribosomal RNA sequencing and reflecting evolutionary relationships rather than structural similarities.

Characteristics of Archaea

Many scientists hypothesize that the Archaea are the closest modern relatives of Earth's first living cells. Called "universal ancestors," these are the cells from which all other life is believed to have evolved. This hypothesis is based on two types of evidence. Genetic analyses indicate that the Archaea domain branches off of the phylogenetic tree at a point that is closest to the tree's root. Furthermore, it has been observed that many of the Archaea prefer to live in extremes of temperature, salt concentration, and pH—environmental conditions thought to be similar to those found on Earth over 3.5 billion years ago, when life first originated.

The Archaea share certain characteristics with Bacteria, others with Eukarya, and have some characteristics that are unique. For example, cells of the Archaea are structurally more similar to Bacteria, live predominantly as single cells, and have cell walls, although the walls do not contain the complex material called peptidoglycan that is a signature molecule of the Bacteria. While some Eukarya have cell walls, it is not a universal characteristic of that domain, and the Eukarya walls are composed of chitin or cellulose, neither of which occurs in cell walls of Archaea or Bacteria. Like the Bacteria, the Archaea lack a membrane-enclosed nucleus and their DNA exists in a circular form. On the other hand, their DNA is associated with histones, a characteristic of Eukarya, and their cell machinery (such as protein-



Evolutionary relatedness of the three domains of life. Distance along the tree is proportional to time since divergence from a common ancestor.

synthesizing enzymes and RNA polymerases) more closely resembles that found in the Eucarya. The lipids that comprise their membranes are unique, resembling neither the Bacteria nor the Eucarya.

Certain members of the Archaea are able to produce methane gas, another unique characteristic. Methane is one of the most important greenhouse gases. An Italian scientist named Alessandro Volta first discovered it as a type of “combustible air” over two hundred years ago. He trapped gas from marsh sediments and showed that it was flammable long before we knew that it was produced by members of the Archaea that lived in salt marsh sediment. Other important habitats for Archaea with this unique ability include the digestive tracts of animals and sewage sludge digesters.

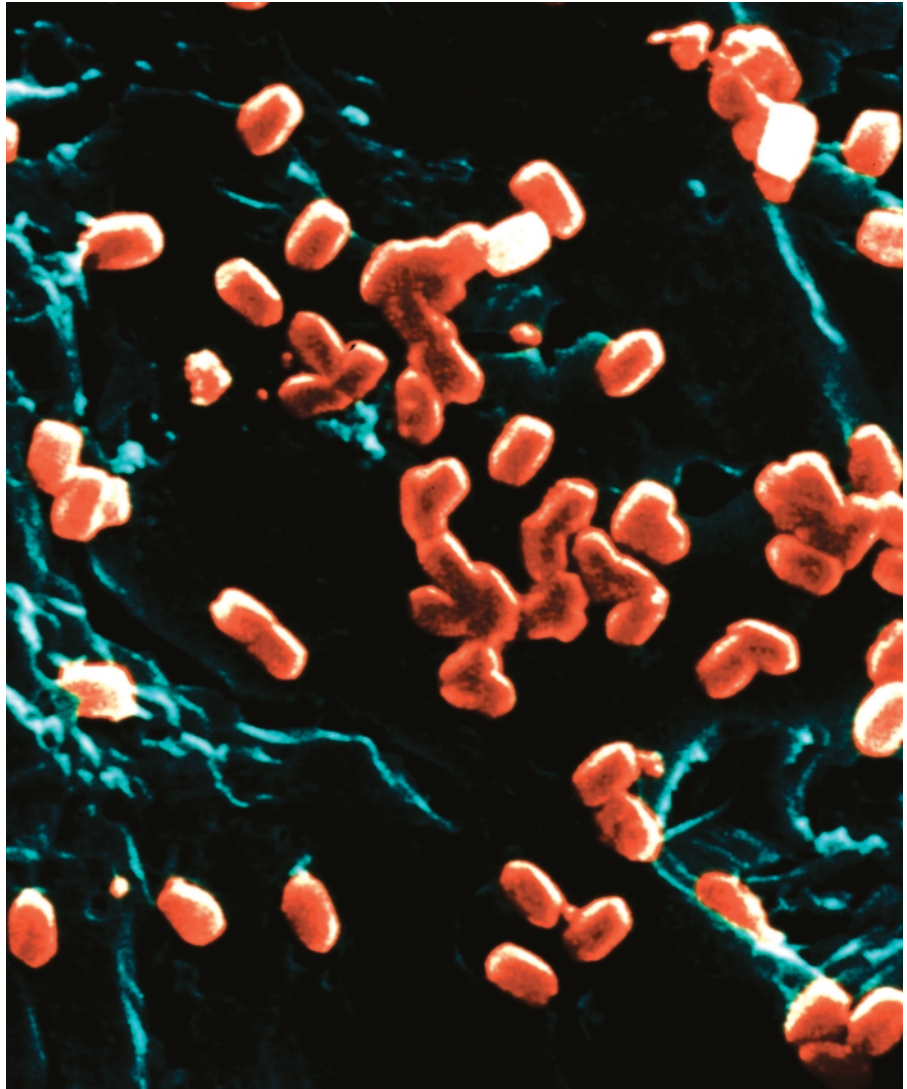
Thriving in Environmental Extremes

The ability of many members of the Archaea to thrive in environmental conditions that we would find extreme is perhaps one of their most fascinating characteristics. There are genus like *Halobacterium*, which inhabit extremely salty environments, such as the Great Salt Lake in Utah and the Dead Sea in Israel. The salt concentration in these lakes is at least ten times that of seawater. Still other lakes, like Lake Magadi in Kenya, are not only extremely salty, but are also extremely alkaline, with pH values as high as 10 or 12. Archaea can be found even here, and their names reflect their habitat: *Natronobacterium*, *Natronosomonas*, and *Natronococcus* (“natro” means “salt”). The reddish-purple color sometimes seen in seawater-evaporating ponds, where solar salt is prepared, is the result of the growth of red-pigmented Archaea.

Extremes of temperature offer no challenge to certain members of the Archaea. A number of species, in fact, require temperatures over 80 °C in order to grow. Some live quite happily in the superheated outflow of geothermal power plants. Others thrive in the conditions of extreme acidity and temperature found in sulfur-rich, acidic hot springs like those in Yellowstone National Park, in the United States.

The high-salt density of the Dead Sea makes it difficult for humans to swim in its waters. Human bodies are much more buoyant in the Dead Sea waters than in fresh water. It is possible to lay back in the water—floating as if on an air mattress.

Archaea bacteria are also known as “extremophiles,” thanks to their ability to survive in extreme environments such as very hot and very cold climates.



Archaea also populate the areas surrounding deep-sea vents, underwater volcanoes that form when the earth’s crust opens along the ocean floor’s spreading centers. The deep-sea vents have the hottest temperatures at which any living organism has been found. As of 2001, the current record for heat tolerance belonged to *Pyrolobus fumarii*, which can grow in water at a maximum temperature of 113 °C, well above boiling. At the opposite extreme, Archaea are among the few organisms found in the frigid waters of the Antarctic.

Value to Industry and Research

As a consequence of their ability to thrive in extreme conditions, the Archaea have become increasingly valuable. For example, the DNA polymerase of *Thermus aquaticus*, an Archaea found in the Yellowstone hot springs, is a heat tolerant enzyme that is crucially important in modern molecular biology laboratories, because of its use in the polymerase chain reaction. Archaea have also become important for commercial purposes. Their enzymes, sometimes called extremozymes, have made their way into

laundry detergent, for example, where they digest proteins and lipids in hot water or cold, and in extremely alkaline conditions, thus helping to remove life's little messes. SEE ALSO CELL, EUKARYOTIC; EUBACTERIA; POLYMERASE CHAIN REACTION; RIBOSOME.

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Attention Deficit Hyperactivity Disorder

Attention deficit/hyperactivity disorder (ADHD) is a condition characterized by inattention and/or impulsivity and hyperactivity that begins in children prior to the age of seven. Their inattention leads to daydreaming, distractibility, and difficulties sustaining effort on a single task for a prolonged period of time. Their impulsivity disrupts classrooms and creates problems with peers, as they blurt out answers, interrupt others, or shift from schoolwork to inappropriate activities. Their hyperactivity is frustrating to those around them and poorly tolerated at school. Children with ADHD show academic underachievement and conduct problems. As they grow older, they are at risk for low self-esteem, poor peer relationships, conflict with parents, delinquency, smoking, and substance abuse.

Course, Prevalence, and Treatment

Although the longitudinal course of this condition and its prevalence in adulthood have been sources of controversy, a growing literature has documented the persistence of ADHD into adulthood, with about two-thirds of ADHD children continuing to experience impairing symptoms of the disorder though adulthood. Over time, symptoms of hyperactivity and impulsivity are more likely to diminish compared with symptoms of inattention.

Prevalence studies from North America, Europe, and Asia show that ADHD affects about 5 percent of the population. The impact of the disorder on society, in terms of financial cost, stress to families, and disruption in schools and workplaces, is enormous. Although current treatments for the disorder are not 100 percent effective, clinical trials have shown that stimulant medications, such as methylphenidate and amphetamine, relieve symptoms and lessen adverse outcomes, while showing few adverse side effects. Because these medicines increase the availability of the **neurotransmitter** dopamine in the brain, dysregulation of dopamine systems has been a primary candidate for the **pathophysiology** of ADHD. But drugs like desipramine and alomoxeline, which have their effects on other brain systems, also exert strong anti-ADHD effects. This suggests that dysregulation of dopamine systems cannot completely explain the pathophysiology of ADHD.

neurotransmitter molecule released by one neuron to stimulate or inhibit a neuron or other cell

pathophysiology disease process



neuroimaging techniques for making images of the brain

frontal lobe one part of the forward section of the brain, responsible for planning, abstraction, and aspects of personality

etiology causation of disease, or the study of causation

bipolar disorders psychiatric diseases characterized by alternating mania and depression

Neuropsychological and **neuroimaging** studies provide converging evidence for the hypothesis that brain dysfunction causes the symptoms of ADHD. Neuropsychological tests show many ADHD patients to have deficits in the executive functions needed for organizing, planning, sequencing, and inhibiting behaviors. These performance deficits are similar to, albeit milder than, the deficits seen among patients with **frontal lobe** disorders. Several structural and functional neuroimaging studies implicate networks of regions throughout the brain, not just in the frontal lobes.

The Genetic Epidemiology of ADHD

Family, twin, and adoption studies provide strong support for the idea that genes influence the **etiology** of ADHD. Family studies find the parents and siblings of ADHD children to have a five-fold increase in the risk for ADHD. Children of ADHD adults have a ten-fold increase in risk, which has led to the idea that persistent cases of ADHD may have a stronger genetic component. Consistent with a genetic theory of ADHD, second-degree relatives (such as cousins) are at increased risk for the disorder but their risk is lower than that seen in first-degree relatives.

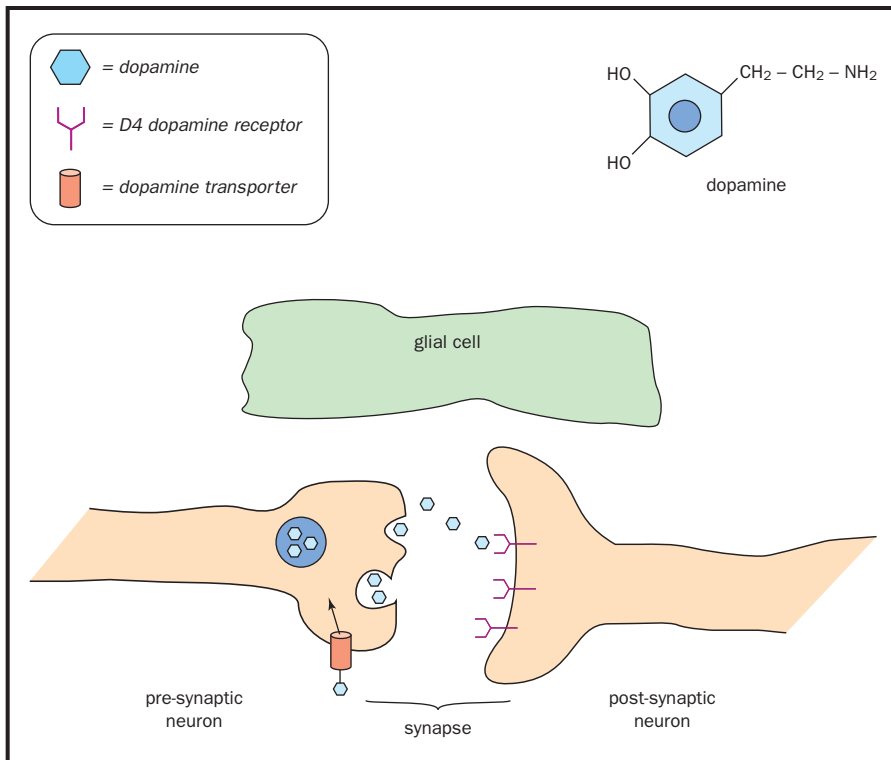
Family studies have provided evidence for the genetic heterogeneity of ADHD. Studies that systematically assess other psychiatric disorders suggest that ADHD and major depression often occur together in families; that ADHD children with conduct and/or **bipolar disorders** might be a distinct familial subtype of ADHD; and that ADHD is familial independent from anxiety disorders and learning disabilities. It may therefore be appropriate to divide ADH children into those with and those without conduct and bipolar disorders, thus forming more familially homogeneous subgroups. In contrast, major depression may be a nonspecific manifestation of different ADHD subforms.

Several twin studies have provided evidence of genetic influence on hyperactive and inattentive symptoms. An early study found the heritability of hyperactivity to be 64 percent. A study of ADHD in twins who also had reading disabilities reported the heritability of attention-related behaviors to be 98 percent. All twin studies considered together suggest that the heritability of ADHD is about 70 percent, which makes it one of the most heritable of psychiatric disorders.

Adoption studies also implicate genes in the etiology of ADHD. Two early studies found that the adoptive relatives of hyperactive children were less likely to be hyperactive or have associated conditions than the biological relatives. Biological relatives of hyperactive children also performed more poorly on standardized measures of attention than did adoptive relatives. A study using the contemporary definition of ADHD found that biological, not adoptive, relationships account for the transmission of ADHD.

The Molecular Genetics of ADHD

Molecular genetic studies have already implicated several genes as mediating the susceptibility to ADHD. Researchers have examined candidate genes in dopamine pathways because animal models, theoretical considerations, and the effectiveness of stimulant treatment implicate dopaminergic dysfunction in the pathophysiology of this disorder. Dopamine is a neural trans-



Dopamine is a neurotransmitter in the brain and is used in neural pathways involved in both movement control and pleasure/reward systems. Dopamine released by one neuron crosses the synapse to stimulate the adjacent neuron. It is broken down by glial cells.

mitter in the brain used in both movement control and pleasure/reward systems. In its simplest form, the dopamine hypothesis holds that excess clearance of dopamine between neurons may contribute to ADHD.

Many studies have focused on the D4 dopamine receptor gene (DRD4) which encodes a protein receptor that mediates the post-synaptic action of dopamine. A meta-analysis of these studies showed a small but statistically significant association, which could not be accounted for by any single study or by publication biases. Although the nature of the mutations in DRD4 have not been conclusively described, a version of the gene known as the 7-repeat allele has generated much interest because this allele causes a blunted response to dopamine and has been implicated in novelty seeking, a personality trait of many ADHD patients.

Several authors have reported an association between ADHD and a particular allele of the dopamine transporter (DAT) gene. This finding has been replicated by some, but not all studies. The link between the DAT gene and ADHD is further supported by a study that relates this gene to poor methylphenidate response in humans, a “knockout” mouse study showing that its elimination leads to hyperactivity in mice, and two molecular neuroimaging studies that found elevated DAT density in the **striatum** of ADHD adults.

Molecular genetics studies of ADHD have also targeted other genes that are related to the dopamine system. Four studies have examined the Catechol-O-Methyltransferase (COMT) gene, whose protein product breaks down dopamine and norepinephrine. Although one study found ADHD was associated with the high-activity form of COMT, three others could not replicate the finding. Other candidate genes that show promising results for ADHD are the D5 dopamine receptor gene and the serotonin

striatum part of the midbrain

1B receptor. This latter finding is intriguing because, although serotonergic medicines do not help ADHD symptoms, these systems have been implicated in animal models of the disorder. SEE ALSO BEHAVIOR; DISEASE, GENETICS OF; PSYCHIATRIC DISORDERS; TWINS.

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Attorney

Attorneys involved with genetics include criminal prosecutors (district attorneys), public defenders, environmental lawyers, family lawyers, and patent attorneys. Genetics is relevant in the areas of identification of suspects and victims, identification of illegal goods (for example, items that involve the killing of endangered animals), environmental monitoring for harmful microorganisms, parentage determinations, and the patenting of genetic materials.

While all of these different types of lawyers may need to be somewhat familiar with the fundamentals of genetics, attorneys who work on gene patents must be very familiar with both genetics and biochemistry, as well as with patent law. The majority of these patent attorneys specialize in biotechnology. Most biotechnology patent attorneys have advanced degrees, with many having Ph.D.s in genetics, microbiology, molecular biology, biochemistry, or related fields. In addition to having a strong science background, patent attorneys must be licensed to practice law in at least one state, and must pass a registration examination administered by the U.S. Patent and Trademark Office (USPTO).

Patent agents, as well as patent attorneys, can represent inventors before the USPTO. Patent agents have strong science backgrounds and must pass the USPTO registration examination, but are not licensed to practice law in any state. In addition to having very strong science backgrounds and the ability to work closely with inventors, patent attorneys and agents must enjoy reading complex scientific literature and be proficient at scientific writing. A large portion of the job involves writing scientific documents in the form of patent applications. Thus, people who enjoy reading and writing about scientific topics are well suited to the profession.

Patent attorneys and agents typically work in law firms, private companies, the U.S. Patent and Trademark Office (patent examiners), or in the technology transfer offices of universities and public institutions such as hospitals and research facilities, although some work out of their homes as solo practitioners. Patent attorneys and agents often know about ground breaking developments long before the public or others in their fields. This makes the career very interesting, exciting, and enjoyable. However, because of

client confidentiality, patent attorneys and agents are required to keep these developments secret until the information is made public by the client or a patent is issued.

While some patent attorneys only draft patent applications and work with patent examiners, others work as litigators, patent law experts, law school professors, or trademark lawyers. For complicated cases, patent attorneys with an education in genetics are very helpful in explaining the technology to the judge or jury. Although most patent attorneys work on patent cases in courts, their expertise may also be called upon in criminal cases, when assistance is needed to analyze and explain complex sciences such as genetics and molecular biology.

Salaries differ widely among patent attorneys and agents who work in law firms, companies, and at universities. As of 2000, new Ph.D.-level patent attorneys could expect to earn at least \$100,000 per year, while new Ph.D.-level patent agents could expect to earn at least \$75,000 per year. However, some patent attorneys could earn well over \$500,000 per year. SEE ALSO LEGAL ISSUES; PATENTING GENES.

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Automated Sequencer

The process of determining the order of nucleotides (A, C, G, and T) along a DNA strand is called DNA sequencing. Knowing the **nucleotide** sequence of a gene or region of DNA is important in studying relatedness between species and between individuals and for a better understanding of how genes function. Several techniques have been developed for “reading” the sequence of any particular DNA segment. One of these techniques was developed by Fred Sanger in 1977 and is called the chain termination method (or Sanger method). The essence of the technique is the creation of a set of DNA fragments that match the chain to be sequenced. Each fragment is one nucleotide longer than the last. By determining the identity of the final nucleotide in each fragment, the sequence of the whole chain can be determined.

The chain termination method makes use of special forms of the four nucleotides that, when incorporated at the end of a growing chain during DNA synthesis, stop (terminate) further chain growth. In four separate reactions, each containing a different terminator base (called a dideoxynucleotide), a collection of single-stranded fragments is made. These fragments all differ in length and all end in the dideoxynucleotide added to the particular reaction. **Gel electrophoresis** is then used to separate the fragments according to their length. By knowing which terminator base is associated with which fragment on the gel, the base sequence can be constructed.

The Need for Automated Sequencing

When chain termination sequencing is performed manually, each of four reaction tubes contains a different type of terminator base as well as a

nucleotide the building block of RNA or DNA

gel electrophoresis technique for separation of molecules based on size and charge



primer short nucleotide sequence that helps begin DNA replication

radioactive nucleotide for labeling the DNA fragments as they are made. Each of the four reactions is electrophoresed in a separate lane of a gel, and X-ray film is used to detect the fragments. Using this technique, a dedicated and skilled technician can determine the sequence of as many as 5,000 bases in a week. Demand for the ability to read more sequence in a shorter amount of time, however, led to the development of instruments that could, with the aid of computers, automate the DNA sequencing process.

The first step toward this goal was achieved in 1985, when Leroy Hood at the California Institute of Technology attached fluorescent dyes to the **primer** used in the sequencing reactions; each different color dye (blue, green, yellow, and red) was matched with a different terminator base. He and Michael Hunkapiller from Applied Biosystems, Inc. (ABI) built an instrument, dubbed the ABI Model 370, to read the sequence of the dye-labeled fragments. It was equipped with an argon ion laser for exciting the dyes, a flat gel laid between two glass plates (referred to as a “slab” gel) capable of sixteen-lane electrophoresis, and a Hewlett-Packard Vectra computer boasting 640 megabytes of memory for data analysis.

Using fluorescent dyes, all four sequencing reactions could now be loaded into a single gel lane. As the fragments electrophoresed, the beam of the laser focused at the bottom of the gel made the dye-labeled fragments glow as they passed. The color of each dye-labeled fragment was then interpreted by the computer as a specific base (A if green, C if blue, G if yellow, and T if red). Over 350 bases could be read per lane. With this new automated approach, a technician could read more sequence in a day than could be read manually in an entire week.

Refinements in Automation

Shortly after ABI placed its automated DNA sequencer on the market, the Dupont company introduced its own model, the Genesis 2000. Dupont had also developed a new method of labeling sequencing fragments: attaching the fluorescent dyes to the terminator bases. With this innovation, four separate sequencing reactions were no longer required; the entire sequencing reaction could be accomplished in a single tube. However, Dupont failed to effectively compete in the automated sequencer market and sold the rights to the dye terminator chemistry to ABI.

ABI continued to refine its automated sequencer. More powerful computers, increased gel capacity (to 96 lanes), improvement of the optical systems, enhancement of the chemistry, and the introduction of more sensitive fluorescent dyes increased the reading capacity of the instrument to over 550 bases per lane. The ABI PRISM Model 377 Automated Sequencer, introduced in 1995, incorporated these changes and could read, at maximum capacity, over 19,000 bases in a day. Even at this rate, however, the sequencing of entire genomes, as that of humans (3 billion bases in length), was still not practical. Genome sequencing awaited several further innovations.

Working with the Model 377 Automated Sequencer, a laboratory technician had to pour the slab gels and mount them on the instrument. This process alone was time-consuming and cumbersome. In addition, the technician had to add each sequencing reaction into each individual lane of the gel prior to the run. The MegaBase, developed by Molecular Dynamics, and

the ABI Model 3700 Automated Sequencer, developed by ABI, addressed these limitations by using multiple capillaries, thin, hollow glass tubes filled with a gel polymer.

The ABI PRISM Model 3700 Automated Sequencer, developed with the Hitachi Corporation and having a price tag of \$300,000, uses ninety-six capillaries, each not much wider than a strand of human hair. The capillaries are automatically cleaned and filled with fresh gel polymer between each electrophoresis run. The instrument is also equipped with a robot arm that automatically loads the sequencing reactions into the capillaries, greatly decreasing the amount of human labor required for its operation. The Model 3700 Automated Sequencer can read over 400,000 bases in a day, a greater than twenty-fold increase over the maximum capacity of the Model 377. Beginning in September 1999 and using 300 of these instruments, the Celera Corporation had sequenced the entire human genome five times over within four months. SEE ALSO CYCLE SEQUENCING; GEL ELECTROPHORESIS; NUCLEOTIDE; SANGER, FRED; SEQUENCING DNA.

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Bacteria See *Eubacteria*, *Archaea*

Balanced Polymorphism

Balanced polymorphism is a situation in which two different versions of a gene are maintained in a population of organisms because individuals carrying both versions are better able to survive than those who have two copies of either version alone. The evolutionary process that maintains the two versions over time is called balancing selection.

Genes are carried on chromosomes. Different versions of a gene are called alleles. The standard allele found in a population is referred to as the wild-type allele. Most plants and animals have at least two copies of each chromosome, one inherited from each parent. The copies of the genes found on these **homologous** chromosomes may be identical or different; that is, the organism may carry two copies of one allele, or one each of two different alleles. In the first case, the organism is called homozygous for that gene, and, in the second, it is called heterozygous.

Alleles differ from each other in their sequence of **nucleotides**, which may change the structure and function of the protein the gene codes for. Because of this, different alleles may have different effects on an organism's appearance or ability to survive. These effects can be helpful, harmful, or neutral.

An example of balanced polymorphism can be illustrated with the set of **enzymes** in the liver that act like an assembly line (or, more accurately, a disassembly line) to detoxify poisons and other chemicals. Different alleles for these enzymes can affect how well an organism can protect itself from

homologous similar in structure

nucleotides building blocks of DNA or RNA, composed of sugars, phosphates, and bases

enzymes proteins that control a reaction in a cell

B

exposure to harmful chemicals. An especially active form of a detoxifying enzyme, which is encoded by a specific allele, can cause accumulation of potentially harmful intermediates. If the other allele encodes an enzyme with low activity, the potential for this enzyme to cause harm is lessened, and the benefits of its activity will be felt by the organism. If an individual has two copies of the very active allele or two copies of the low-activity allele, it may not survive well. In the first case, too much enzyme activity will result in high levels of the harmful intermediate, and in the second case, too little enzyme activity will be present for detoxification. Therefore, the best situation for the organism is to have one copy of each allele. Because of this, both copies are maintained in the population.

The effects of alleles and whether they are maintained in a population can be influenced by the environment. A classic case of balanced polymorphism in humans that is influenced by the environment is the sickle-cell allele of the β -globin gene. This gene forms part of hemoglobin, which carries oxygen in red blood cells.

Individuals who have two copies of the β -globin sickle-cell allele develop sickle-cell disease and generally do not survive into adulthood without intensive medical care. Individuals with one copy of the β -globin sickle-cell allele and one β -globin wild-type allele have red blood cells that are functional and resistant to the organism that causes malaria. Because individuals with this combination of alleles tend to survive malaria better than those who carry only the wild-type allele, the combination is advantageous to those who live in areas where malaria is present. This is called “heterozygote advantage.” As a result, the beta-globin sickle-cell allele will be maintained along with the wild-type allele in populations exposed to malaria—an example of balancing selection. SEE ALSO ALZHEIMER’S DISEASE; HEMOGLOBINOPATHIES; HETEROZYGOTE ADVANTAGE; POPULATION GENETICS.

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Barr Body See *Mosaicism*

Behavior

What is behavior? A dictionary definition reveals that behavior consists of our activities and actions, especially actions toward one another. As such definitions suggest, many behavioral terms have meaning only in social comparisons: We identify others as contentious, courteous, or conscientious only by their actions in social contexts. A long-standing question in science and in everyday affairs inquires about the causes of individual differences in behavior: Why are some people gregarious extroverts and others timid, shy introverts?

Behavior genetics is a **hybrid** area of science, at the intersection of human genetics and psychology. Its focus is on how genes and environments contribute to differences in behavior. It is a young discipline. A book that

β the Greek letter beta

hybrid combination of two different types